

Thomas Lee

Senior Software Engineer

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Senior Full Stack Engineer experienced in designing and delivering high-performing web applications and scalable APIs for enterprise environments. Deep expertise across frontend and backend technologies, including Java (Spring Boot), .NET, Ruby on Rails, JavaScript frameworks, and modern cloud and DevOps practices. Led the development of real-time analytics platforms and AI-driven healthcare solutions, significantly improving business insights and automated diagnostics. Ready to drive innovation and reliability in software systems by applying advanced engineering skills and a proven ability to solve complex technical challenges.

Work Experience

Software Development Engineer II
Amazon - Atlanta, GA

Feb 2022 - Present

- Architected robust Web APIs, adhering to RESTful principles to streamline integration with third-party systems and internal services with **Java & Spring Frameworks, MySQL**.
- Architected a real-time data analytics platform using **Java & Kotlin** and **Apache Kafka**, processing and analyzing large volumes of streaming data to provide actionable insights to stakeholders.
- Configured and maintained monitoring solutions using **Zabbix** and **Grafana** to ensure high availability and performance of API and microservice environments.
- Developed custom **Zabbix** templates and triggers for API health checks, latency tracking, and resource utilization monitoring across distributed systems.
- Integrated **Zabbix** with **Prometheus** and Alertmanager, creating a unified observability stack for real-time alerting and trend analysis.
- Configured **Kafka** clusters with partitioning, replication, and consumer groups to optimize performance and scalability.
- Designed and deployed cloud-native microservices on **AWS** using services such as **EC2, Lambda, API Gateway, S3, RDS**, and **CloudFront** to build highly scalable, fault-tolerant applications.
- Developed Infrastructure as Code (**IaC**) with Terraform by creating reusable, modular configurations to automate provisioning of VPCs, security groups, load balancers, and **CI/CD** pipelines, reducing deployment time and minimizing configuration errors.
- Implemented multi-environment deployments (dev, staging, production) with **Terraform Cloud/Enterprise**, ensuring consistent and automated infrastructure changes through version-controlled Git workflows.
- Developed and maintained enterprise-grade web applications using **Node.js / NestJS**, and **Ext JS**, ensuring clean architecture and modular design.
- Built unit and integration tests using Jest in conjunction with **NestJS** to ensure application reliability and maintain high-quality code standards.
- Integrated third-party libraries such as Swagger for API documentation, Passport.js for authentication, and TypeORM for database management within a **NestJS** environment.

- Developed Infrastructure as Code (IaC) with Terraform by creating reusable, modular configurations to automate provisioning of VPCs, security groups, load balancers, and CI/CD pipelines, reducing deployment time and minimizing configuration errors.
- Reduced data processing times by 40% through **SAP HANA** database optimization techniques.
- Successfully migrated legacy systems to **SAP HANA**, ensuring minimal downtime and data integrity.
- Worked with cross-functional teams to improve reporting capabilities with HANA-native tools such as **SAP Fiori** and **SAP Analytics Cloud**.
- Designed and developed custom **SAP** reports and dashboards using **SAP BI** and other analytics tools, ensuring business stakeholders received timely and accurate data for decision-making.
- Integrated **Python** services with **Kafka** for producing and consuming real-time event streams, and with **MongoDB** for high-performance data persistence and analytics.
- Implemented end-to-end API testing frameworks leveraging **Postman**, **JUnit**, and **pytest** to ensure consistent quality and high code coverage across environments.
- Documented APIs using **Swagger/OpenAPI** and Markdown, improving internal knowledge sharing and external developer onboarding.
- Designed, developed, and maintained enterprise-grade **RESTful APIs** and microservices enabling seamless integrations between internal systems and third-party partners.
- Applied data analysis and visualization techniques using libraries like **Pandas**, **NumPy**, and **Matplotlib** to support business insights and reporting.
- Utilized **MongoDB** as a primary data store for structured and semi-structured data, leveraging indexing strategies and aggregation pipelines to achieve high query performance.
- Implemented change data capture (CDC) using **Kafka Connect** and **MongoDB** connectors to synchronize data between operational and analytical systems in real time.
- Executed the migration of a monolithic Rails application, transitioning the front end to a **React** and **Next.js** framework while developing APIs in **Ruby on Rails**. Accomplished in three weeks, resulting in the successful launch of the revamped application.
- Crafted APIs utilizing capabilities of **Ruby on Rails**, ensuring reliable and efficient communication between various components, which improved system performance and reliability.
- Built and launched new features using **Ruby on Rails** and **React**, improving user experience and increasing customer engagement by submitting well-documented **GitHub** PRs.
- Used **GitHub Copilot** and **ChatGPT** to accelerate code reviews, generate unit tests, and prototype new service endpoints, reducing development time significantly.
- Achieved **Angular** authentication for the application and stored generated token so that token can be sent with each request to secure resources in Web API.
- Worked on **Angular** Directives, Dependency Injection, Services, Routing, pipes, Animations, Forms, AOT, Lazy Loading, RXJS.
- Achieved over 90% test coverage by meticulously designing unit tests and using **SonarQube** with **Docker** to monitor code quality.
- Implemented migrations for a **.Net** legacy application to **Java Spring Boot** and version control for **PostgreSQL** schemas using tools like **Liquibase** and **Flyway**, ensuring seamless transitions between development, staging, and production environments.
- Spearheaded the development of the AI-powered ASCO Docs platform, utilizing **.NET 6.0**, **ASP.NET Core 6**, **C#.Net** and Entity Framework Core for efficient document generation.
- Implemented **React.js** 18 for building seamless user interfaces, ensuring a responsive, intuitive front-end experience.
- Fostered collaboration and improvement within the agile development team to enhance overall project outcomes.

- Analyzed large-scale healthcare datasets using clinical data analytics tools to uncover patterns and trends that improved patient diagnosis decisions. (**NumPy, Pandas, Power BI, R, MySQL**) (**Clinical Data Analysis & Insights**)
- Developed AI-based models using **Python** and deep learning (**TensorFlow, PyTorch, Keras**) to accurately detect early signs of diseases such as cancer, cardiovascular conditions, and diabetes from medical images (e.g., X-rays, MRIs). (**AI-Based Diagnostic Models**)
- Developed and implemented AI algorithms (such as convolutional neural networks and natural language processing) for medical image analysis, significantly enhancing the detection and diagnosis of anomalies in radiology and pathology slides. (**AI in Medical Imaging**)
- Enhanced LegalZoom's document platform with **TypeScript** and **React.js**, boosting user engagement by 20% and increasing customer satisfaction scores.
- Designed and implemented scalable backend services using **Node.js, Express**, and **TypeScript**, improving API response times by 30%.
- Created and optimized custom middleware in **NestJS** for handling authentication, logging, and request transformations, enhancing request lifecycle management.
- Optimized cloud solutions using **Microsoft Azure**, including **Azure DevOps** for CI/CD pipelines and Azure Key Vault for secure data management based **Python & FastAPI**.
- Customized **SAP** modules (such as MM, SD, FI/CO, HR) to meet specific business requirements, optimizing workflows, user interfaces, and business processes, which improved operational efficiency and user satisfaction.
- Designed and developed intuitive **SAP UI5** and **Fiori** applications that improved user experience, increased user productivity, and enabled seamless mobile access through modern, responsive interfaces.
- Refactored legacy **APIs** to improve performance, reduce technical debt, and enhance developer experience (DX) through improved documentation and standardization.
- Developed SDKs and reusable **API** components for internal and partner teams, improving onboarding time and reducing integration complexity.
- Integrated Hibernate DAOs with Spring for **Dependency Injection**, enhancing application modularity and reducing code complexity.
- Led code optimization efforts, employing unit testing frameworks like **xUnit** to ensure high code coverage and maintainability.
- Developed and maintained **Kafka** producers and consumers to facilitate seamless integration between microservices, ensuring data reliability and fault tolerance.
- Designed and implemented backend systems with **Go & Gin**, integrating RESTful APIs and microservices to increase scalability, reliability, and overall system performance.
- Integrated REST APIs and improved client-server interactions with **Redux**, resulting in faster page loads and a smoother user experience.
- **Designed and implemented ETL processes** using **SSIS** to extract data from various source systems (SQL Server, Oracle, flat files) and load it into a centralized **data warehouse** for reporting and analysis.
- Managed the client-side aspects of continuous integration through **Jenkins** and maintained **Git** repositories.
- Implemented observability and monitoring using **Grafana** to capture metrics, logs, and distributed traces for proactive issue detection.

Software Engineer I (Fullstack Engineer)
Pegasystems - Alpharetta, GA

Jun 2017 - Mar 2020

- **Increased Forecasting Accuracy:** Developed AI models that improved forecasting accuracy by over 30%, reducing energy waste and enabling more efficient grid management.

- Applied **AI algorithms** to **real-time climate data** (e.g., cloud cover, temperature, wind speed) to optimize the efficiency of **solar power plants**.
- Worked with **energy optimization models** to predict and adjust system performance in real-time, enabling solar plants to make dynamic adjustments based on changing weather conditions, increasing overall energy efficiency by up to 25%.
- Initiated and delivered Lifetouch's Phoenix application using **.NET Core, Python, and AngularJS**, increasing application stability by 30% and boosting user satisfaction scores.
- Refactored legacy **Java 11** code to **Kotlin**, reducing boilerplate and making the code more concise, leading to a 30% improvement in development speed and reducing technical debt.
- Created cloud-native applications on **AWS**, using **Kotlin** in combination with cloud technologies like **EC2, Lambda, and DynamoDB**, ensuring cost-efficient scalability.
- Engineered and executed **MySQL & PostgreSQL** queries that enhanced database performance, achieving a 50% decrease in query response time while ensuring data integrity through routine audits of over 1 million records.
- Developed and maintained web applications using **Node.js** and **Java 11**, focusing on building efficient back-end services and RESTful APIs with **Microservice** Architecture.
- Leveraged problem solving abilities to design and implement innovative software solutions increasing overall system efficiency by 40%.
- Engaged in collaborative code review sessions, maintaining coding quality and fostering developer teamwork.
- Incorporated **SQL scripts** and **Stored Procedures** within **SSIS** for advanced data transformation and integration, optimizing query performance and reducing execution times.
- Collaborated with the product team to define user stories and technical specifications, ensuring user needs and business goals were met and directly improving user experience on the **MERN** project.
- Utilized **NestJS** modules to architect clean, modular applications, enhancing code maintainability and scalability by promoting separation of concerns.
- Implemented dependency injection in **NestJS** to streamline service management and ensure testability, contributing to a more efficient development cycle.
- Managed a team of 5 frontend developers, providing mentorship and conducting code reviews to ensure adherence to best practices and coding standards. Directed the migration of a high revenue enterprise application from **Ember.js** to **React.js**.

Software Development Intern *OpenSpan - Alpharetta, GA*

Jan 2016 - May 2016

- Enhanced the e-commerce platform by developing new features with **ASP.NET** and **MVC** architecture, resulting in a smoother user experience and increased customer satisfaction.
- Improved data processing workflows using **SQL Server**, leading to a 30% reduction in query times.
- Developed complex landing pages and forms with **HTML5/CSS3** to support dynamic layout and responsive design standards.
- Employed **jQuery** and **Bootstrap** to enrich the user interface and improve customer experience.
- Supported continuous integration and deployment practices, helping reduce deployment errors and speed up software releases.

Core Skills

- JavaScript/TypeScript, Node.js, C/C++/C#, Java/Go/Python, Ruby, Kotlin, PHP

- Spring/Spring MVC/Spring Boot/Spring Cloud, Django, Flask, Fast API, Express, Next.js, Ember.js, React.js, Redux, Angular, Vue.js, CI, Laravel, Symfony
- JPA, Hibernate, MyBatis/iBatis, Maven/Gradle, Terraform, Ansible, RabbitMQ, Kafka, Gin
- Solidity, Web3.js, Ethers.js, Truffle, Hardhat, IPFS, NFT(ERC-721, ERC-1155), DAO, DeFi(DEX, Smart Contracts), Wagmi
- MUI, Ant Design, Headless UI, Tailwind CSS, Bootstrap, PrimeNG, Chakra UI
- AWS Lambda/EC2/S3/Cloud, Google Cloud Platform, Azure
- Jenkins, GitLab, CircleCI, Docker, Kubernetes,
- MySQL, PostgreSQL, MongoDB, Redis
- Profiling, Caching Strategies, Load Balancing, Asynchronous Programming
- Jest, chai, Mockito, JUnit, SonarQube, RestAPI, GraphQL, WebRTC
- Figma, Agile, Scrum, Jira
- TensorFlow, PyTorch, Keras, Pandas, NumPy, SciPy
- Apache Solr, Hadoop, Elastic Search, Spark, Tableau, Power BI

Education

Georgia Institute of Technology
Bachelor's Degree | Computer Science

Sep 2013 - Sep 2017

Certifications

AWS Certified Solutions Architect – Associate

Credential ID DJ905M7KH1V41Y3W (Issued Aug 2020 – Expired Aug 2023)

AWS Certified Data Analytics – Speciality

Credential ID CQZ55C0C3JF4QK9L (Issued Jun 2020 – Expired Jun 2023)

AWS Certified Cloud Practitioner

Credential ID H1913L42NJQQ1NCH (Issued Jan 2020 – Expired Jan 2023)

Codility Golden Award (Niobium 2019)

Codility (Issued Jul 2019 – Expired May 2021)

Codility Silver Award (Molybdenum 2019)

Codility (Issued Jul 2020 – Expired Jul 2021)

AWS Certified Database - Speciality

Credential ID Y7MN7CBBE24Q1R3H (Issued Oct 2021 – Expired Oct 2024)